

11.2 Fundamental Particles

Question Paper

Course	CIE A Level Physics
Section	11. Particle Physics
Topic	11.2 Fundamental Particles
Difficulty	Medium

Time allowed: 10

Score: /5

Percentage: /100

Question 1

Which of the following sentences about antimatter is **incorrect** ?

- A. antimatter is normal matter with an opposite charge
- B. antimatter is only produced in particle accelerators
- C. antimatter destroys matter
- D. equal amounts of antimatter and matter were created during the Big Bang

[1 mark]

Question 2

Fundamental or elementary particles are particles that aren't made up of smaller particles.

What is the most common type of fundamental particle in the universe ?

- A. atom
- B. meson
- C. neutrino
- D. quark

[1 mark]

Question 3

In a nucleus of Tritium, ${}^3_1\text{H}$, how many 'down' quarks are there?

- A. 5
- B. 4
- C. 3
- D. 2

[1 mark]

Question 4

Which row in the table gives the correct quark combination for the particle?

	Particle	Category	Quark combination
A	neutron	baryon	$\bar{u}d$
B	neutron	meson	udd
C	proton	baryon	uud
D	positive pion	meson	$\bar{u}d$

[1 mark]

Question 5

What are the numbers of hadrons, baryons and mesons in an atom of ${}^7_3\text{Li}$?

	Hadrons	Baryons	Mesons
A	7	3	3
B	7	4	4
C	7	7	0
D	10	7	0

[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

- Every fermion has an antimatter equivalent e.g. the proton has a quark composition of uud (up, up, down) and the antiproton has a quark composition of $\bar{u}\bar{u}\bar{d}$ (anti-up, anti-up, anti-down)
 - Antimatter is produced synthetically in particles accelerators for experiments
 - It is also produced naturally through the radioactive decay of potassium 40, cosmic ray interactions in the atmosphere and thunderstorms
 - It has also been detected in the Earth's Van Allen radiation belts
- Therefore, antimatter does not just exist as a result of particle accelerator experiments

A is incorrect because antimatter is normal matter but with an opposite charge, for example, the antimatter particle of an electron, which has a negative charge, is the positron, which has a positive charge – both particles are identical in every other way

C is incorrect because matter and antimatter annihilate each other when they come into contact

D is incorrect because equal amounts of matter and antimatter would have been created in the Big Bang (however, it is still unknown why we live in a universe dominated by matter with almost zero antimatter)

2

The correct answer is **C** because trillions of neutrinos are produced every second in nuclear reactions at the center of stars and in supernova explosions when stars die

A is incorrect because atoms are not fundamental particles as they are made up of quarks and electrons

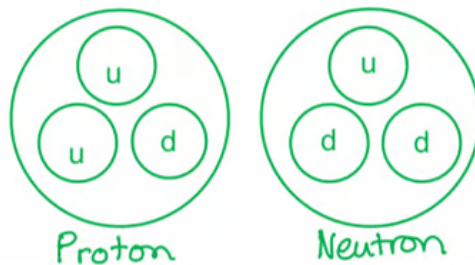
B is incorrect because mesons are not fundamental particles as they are made up of quarks

D is incorrect because while a quark is a fundamental particle, it is not found in as high numbers as the neutrino

3

The correct answer is **A** because:

- Protons and neutrons are composed of two types of quarks held together by gluons:
 - Up quarks
 - Down quarks
- They are both made of three quarks as shown in the diagram

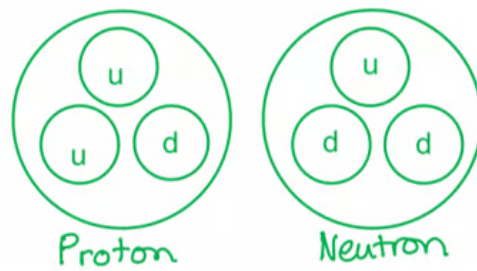


- Tritium is known as hydrogen-3, therefore, it has 2 neutrons and 1 proton
 - Each neutron has two down quarks, making a total of 4
 - The proton has one down quark, giving an overall total of 5
- Therefore, a tritium nucleus contains 5 'down' quarks

4

The correct answer is **C** because:

- There are two types of hadron:
 - Baryons: three quarks or three antiquarks
 - Mesons: a quark and an anti-quark pair
- Protons and neutrons are baryons composed of up and down quarks held together by gluons:
- They are both made of three quarks as shown in the diagram



- Therefore, option **C** is correct

A is incorrect because a neutron has three quarks, two down and one up

B is incorrect because a neutron is not a type of meson; a meson contains one quark and one anti-quark bound by a gluon

D is incorrect because a positive pion is a meson but consists of one u and one $\bar{d}$ quark

5

The correct answer is **C** because :

- Hadrons are quark-based particles, and protons and neutrons are both made of different types of quarks
 - Baryons are a type of hadron
- Lithium contains three protons and four neutrons giving a total of seven hadrons and baryons

A & B are incorrect because a meson contains one quark and one anti-quark bound by a gluon; lithium does not contain any mesons